MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6842

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, high voltage, sharp cutoff

Outline --- 5-2 (EIA) (See Figure 1 for dimensions.)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin no.	1	2	3	4	5	6	7	Cap
Element	G2	H	K, G3	G1	K, G3	H	K, G3	a

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rg1	Ib(ave)	ib	Pp	Pg2
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ma	ma	W	W
Maximum:	6.9	4,000	---	150	±300	0.1	10	100	8	.5
Minimum:	5.7	---	-100	---	---	---	--	---	--	---
TEST CONDITIONS:	6.3	1,500	-1.5	100	---	---	--	---	--	---

GENERAL:

Qualification - Not required

First article testing - Required (See MIL-E-1, appendix F)

FSC 5960

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TABLE I Quality Conformance Inspection.

Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1							
1301	Heater current		---	---	If	135	165	mA
1336	Heater-cathode leakage	Ehk = ±100	1	I	Ihk	---	50	µAdc
1266	Total grid current	(See note 1)	1	I	Ic1	---	-0.3	µAdc
1256	Electrode current (1) (anode)	(See note 1)	1	I	Ib	2.6	4.5	mAdc
1256	Electrode current (screen grid)		1	I	Ic2	---	1.0	mAdc
1306	Transconductance		1	I	Sm	2050	3650	µmhos
	Quality conformance inspection, part 2							
1256	Electrode current (2) (anode) (idling)	Ec1 = -10.0 Vdc	2.5	---	Ibo	---	100	µAdc
1331	Direct-interelectrode capacitance	Unshielded	---	---	Cgp Cin Cout	--- 3.0 1.70	.040 5.0 2.3	pF pF pF
1031	High-frequency vibration	Rp = 2,000 ohms; No voltages	---	(See note 2)	Ep	---	200	mVac
1211	Insulation of electrodes		---	(See note 2)	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
1041	Shock	30° hammer angle	---	(See note 2)	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group D Eb = 3,000 Vdc Ec2 = 100 Vdc RK = 2,500 ohms Ehk = +100 Vdc	---	---	t	500	---	hrs
---	Intermittent life test end points (500 hours):							
1256	Electrode current (1) (anode)		---	---	---	2.0	---	mAdc
1306	Transconductance (change of individual tubes)		---	---	ΔSm t	---	25%	µmhos

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NOTES:

1. This test shall be performed at the conclusion of the holding period.

2. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

3. Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User Activities:
Air Force - 11

Preparing Activity:
Air Force - 85

(Project 5960-3309)

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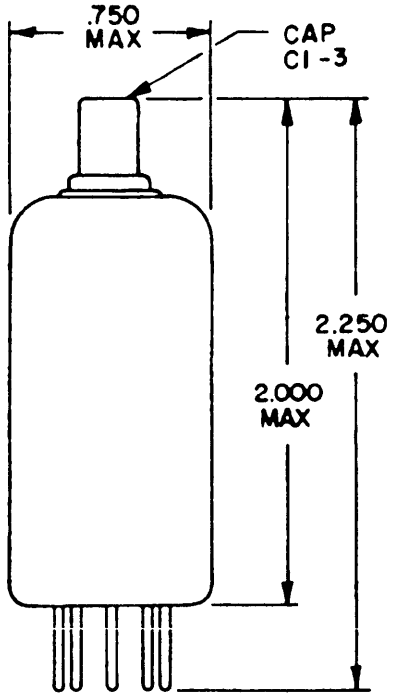


FIGURE 1. Outline drawing for type 6842.

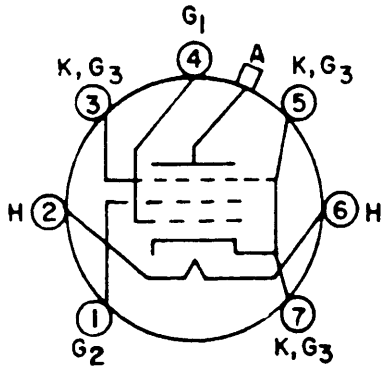


FIGURE 2. Schematic for type 6842.

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